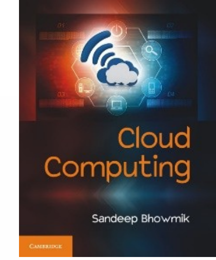
Cloud Computing

Chapter 11

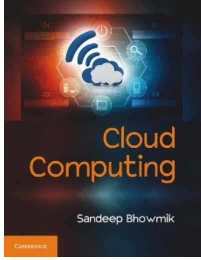
Load Balancing

Cambridge University Press



Introduction

- The dynamic resource provisioned or de-provisioned and horizontal scaling provides elasticity to cloud computing.
- But this flexibility comes with the cost of a lots of implementation complexity.
- Load balancing is the mechanism through which these features are implemented.

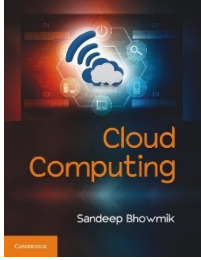


Introduction

- In a distributed computing environment, load balancing is done to proportionately distribute service requests among available resources.

- An efficient load balancing mechanism improves average resource utilization rate, and therefore enhances the overall performance of a system.

- Load balancing is an important ingredient towards building any dynamic, flexible, scalable and robust distributed computing architecture.

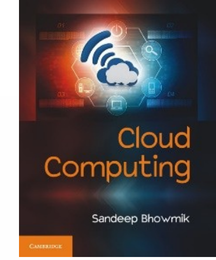


Introduction

- When one component fails for some reason, load balancer can redirect the load to other operating components.

- Thus, the load balancer increases the reliability of the system .

- Load balancing makes a system more productive when additional capacity is induced into a system.

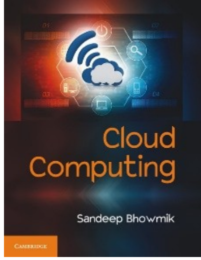


Importance of Load Balancing in Cloud Computing

- A reliable computing system must operate efficiently under varying load conditions.

- The distributed architecture makes load balancing an essential element of cloud to maintain operational efficiency.

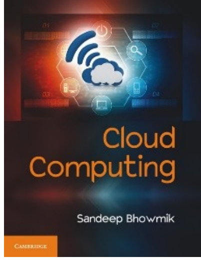
- Load balancing is an essential component for building scalable application architecture.



Importance of Load Balancing in Cloud Computing

- Following points summarize the importance of balancing the load in cloud computing -

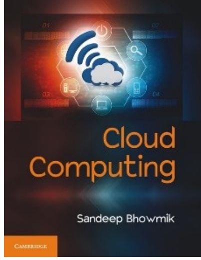
- Load balancing offer architectural flexibility and essential in making a computing architecture scalable.
- It ensures efficient utilization of a pool of similar type resources.
- Efficient resource utilization automatically enhances the performance of the overall system.



Importance of Load Balancing in Cloud Computing

- Following points summarize the importance of balancing the load in cloud computing -

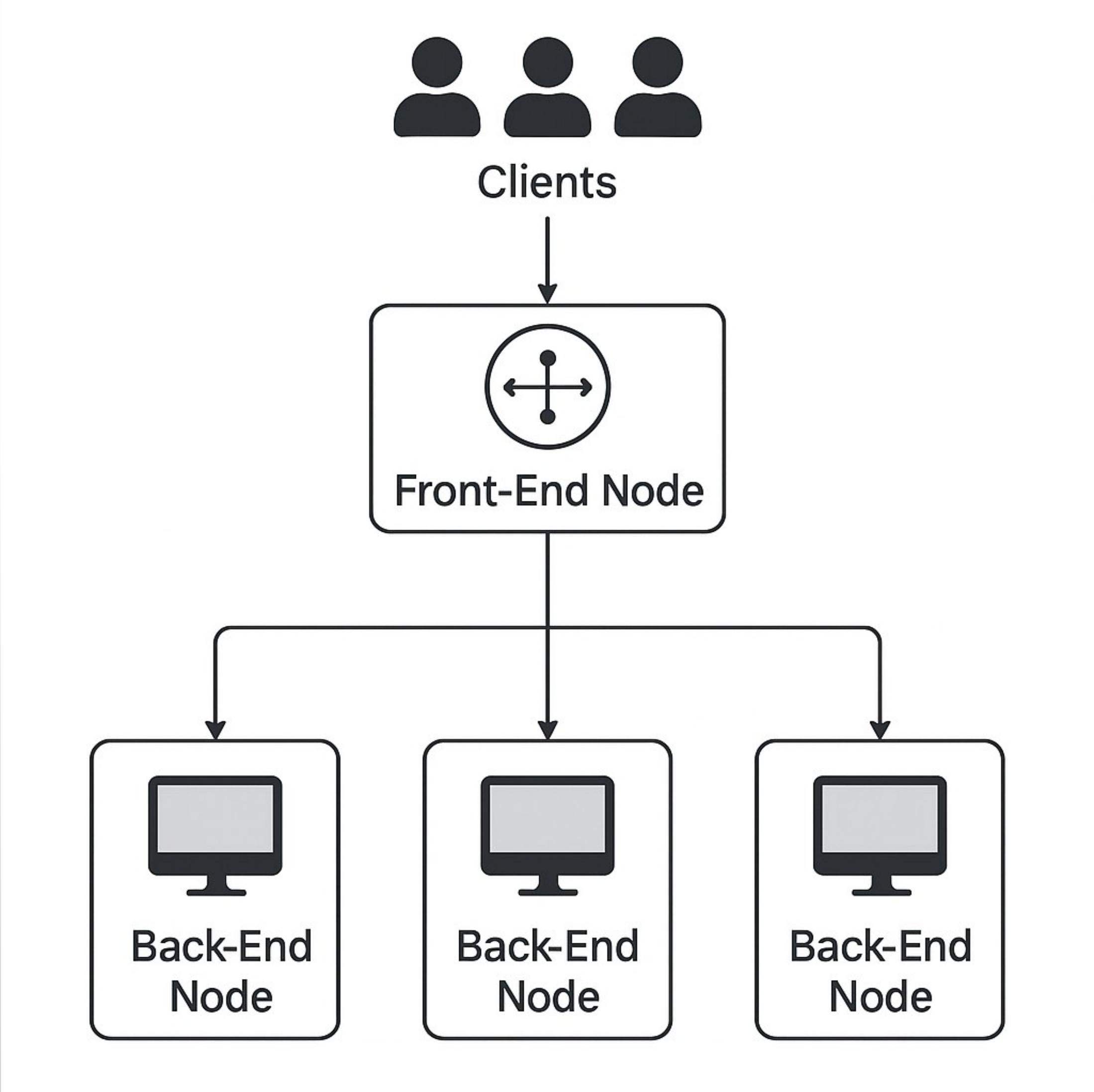
- The technique decouples applications from its physical implementation during execution. This creates a layer of abstraction, which increases application and system security.
- The decoupling of physical resources from direct access of applications also makes the cloud computing system more tolerant in case of any component failure.

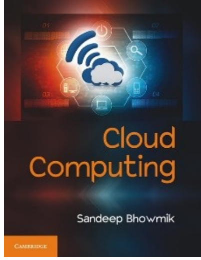


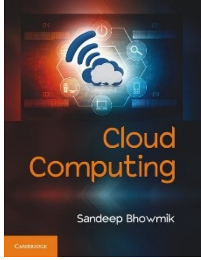
How Load Balancing is done in Cloud

- The load balancer system in a cloud implementation that directly interfaces with clients is called the *front-end node*.
- All the incoming requests first arrive in this front end node at the service provider's end.
- This node then distributes the requests towards appropriate resources for further execution.
- These resources, which are actually virtual machines, are called *back-end nodes*.

Working..


Clients
Front-End Node
Back-End Node
Back-End Node
Back-End Node

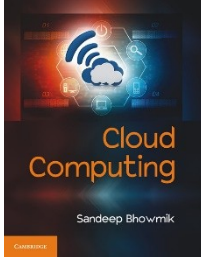

Cloud Computing
Sandeep Bhowmik

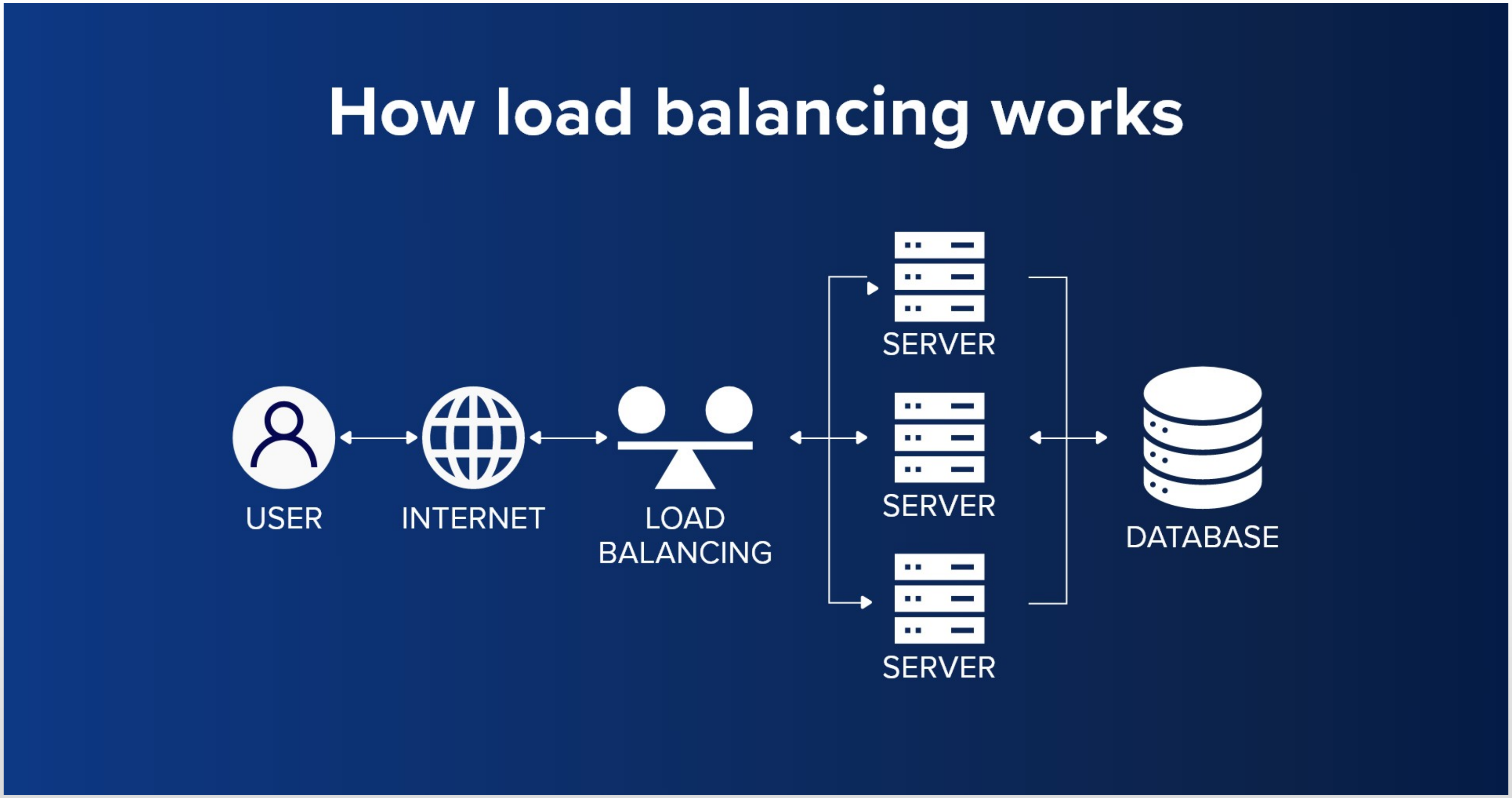


How Load Balancing is done in Cloud

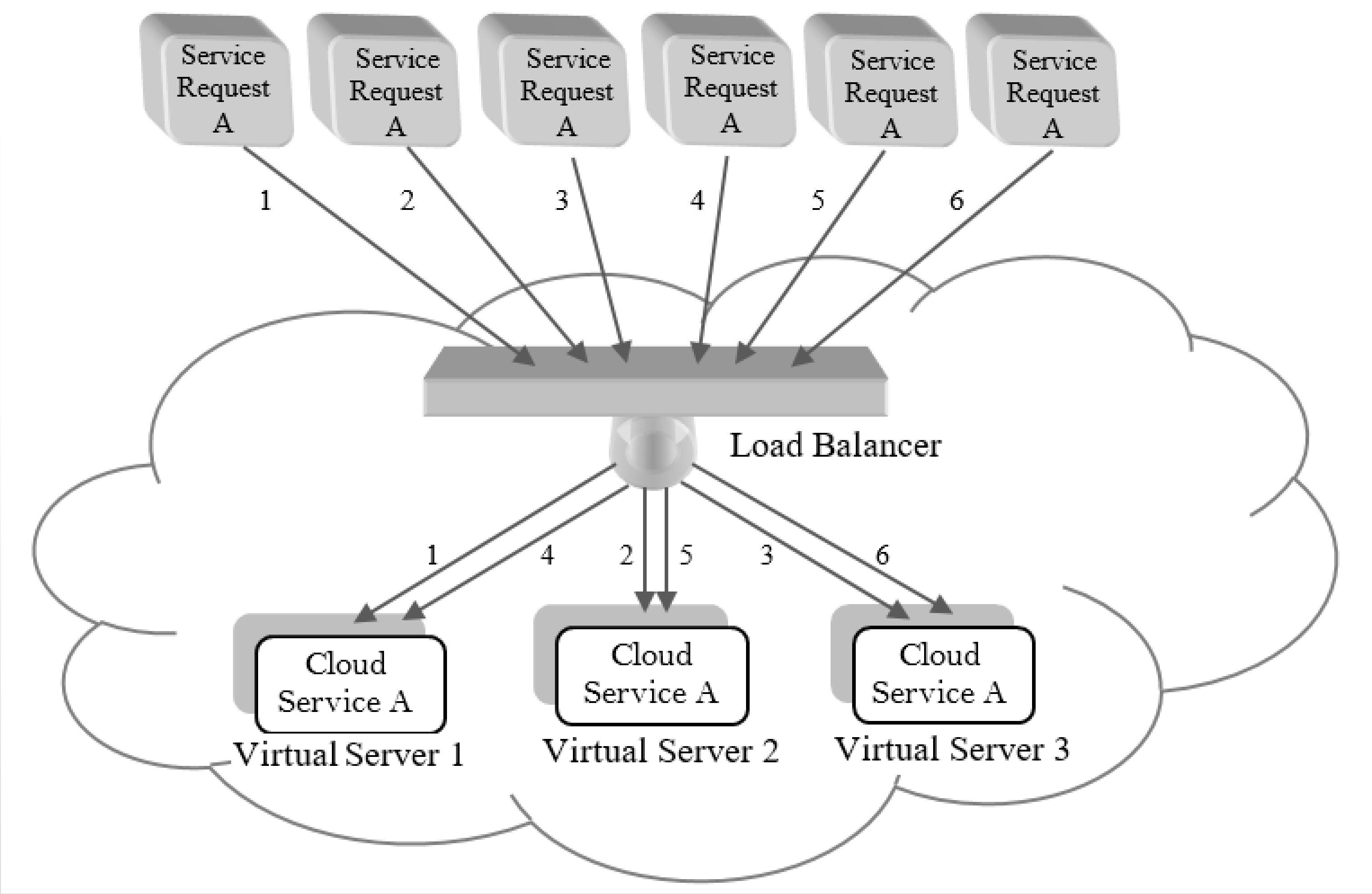
- When the load balancer at the front-end node receives multiple requests for a particular service, it distributes those requests among available virtual servers.
- This distribution happens based on some defined scheduling algorithm.
- This scheduling happens depending on some policy, so that all the virtual servers stay evenly loaded.

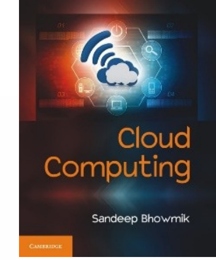
How load balancing works ….?





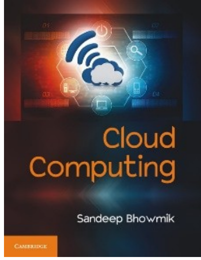
How Load Balancing is done in Cloud





How Load Balancing is done in Cloud

- The use of multiple similar resources with load balancing technique, instead of a single powerful resource, increases system's reliability through redundancy.



How Load Balancing is done in Cloud

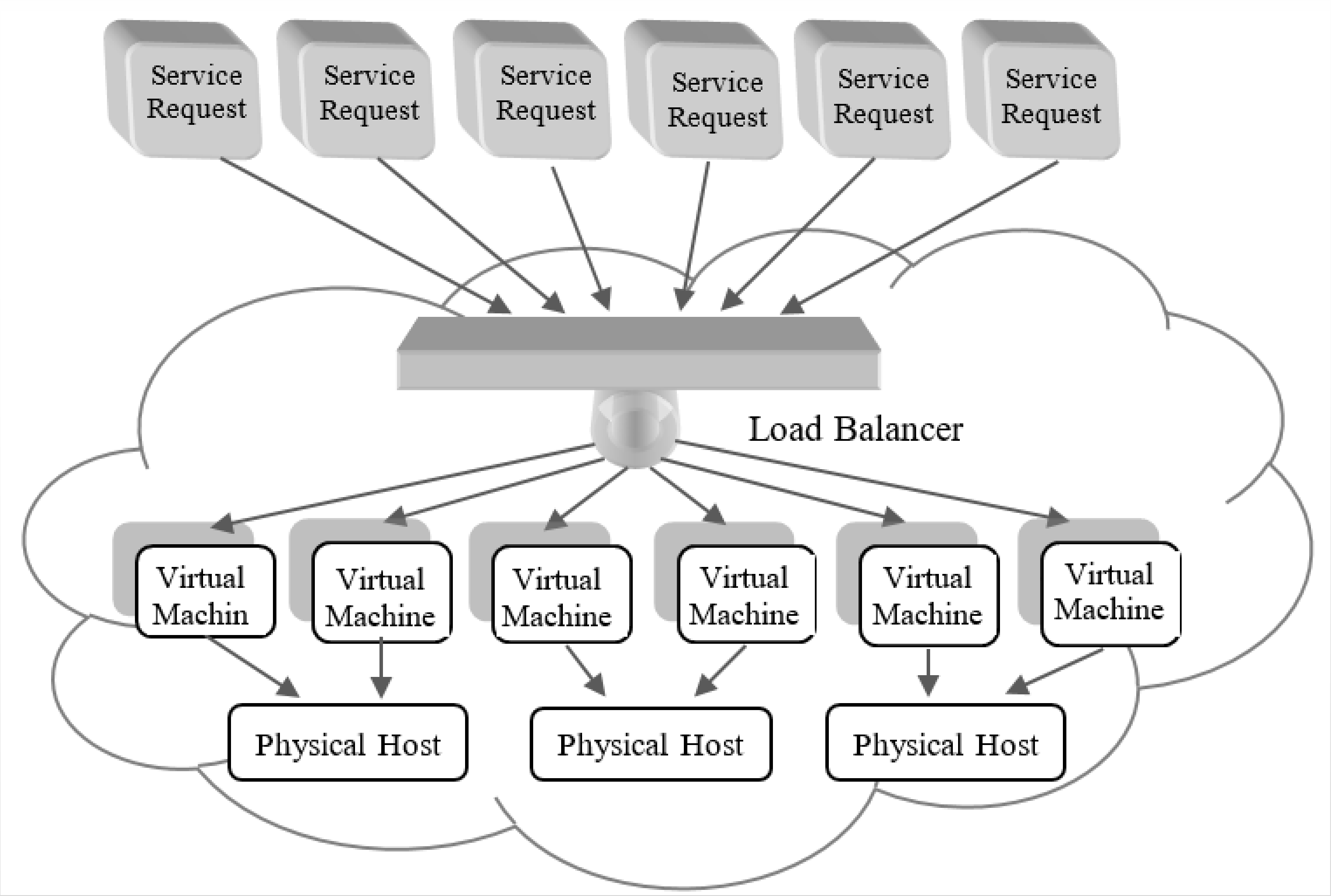
- **Two Levels of Balancing**

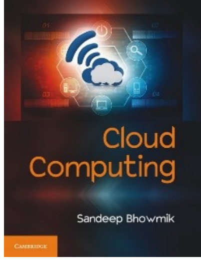
- The assignment and execution of application requests at the back-end-nodes happen in two phases -

- In the first step, a service request is analyzed and assigned to an appropriate virtual machine instance, which is called *VM provisioning*.

- In the second step, these requests are mapped and scheduled onto actual physical resources, and this is known as *resource provisioning*.

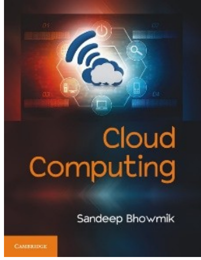
How Load Balancing is done in Cloud





Goals of Load Balancing

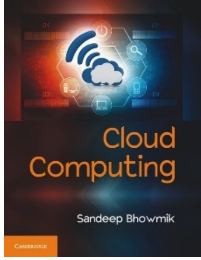
- Improve system performance
- Maximize fault tolerance
- Accommodate scaling
- Ensure availability of applications all the time



Categories of Load Balancing

- For efficient resource utilization as well as effective task execution, a load balancer should have knowledge about the engagements of resources while assigning tasks to resources.

- Depending upon the use of knowledgebase, there are two categories of load balancing technique – static and dynamic.



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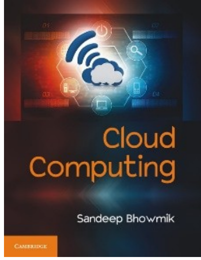


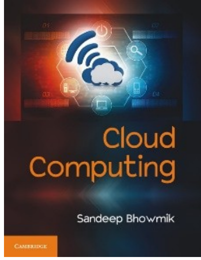
Categories of Load Balancing

- **Static Approach**

- This approach doesn't consider the current load state of the systems.

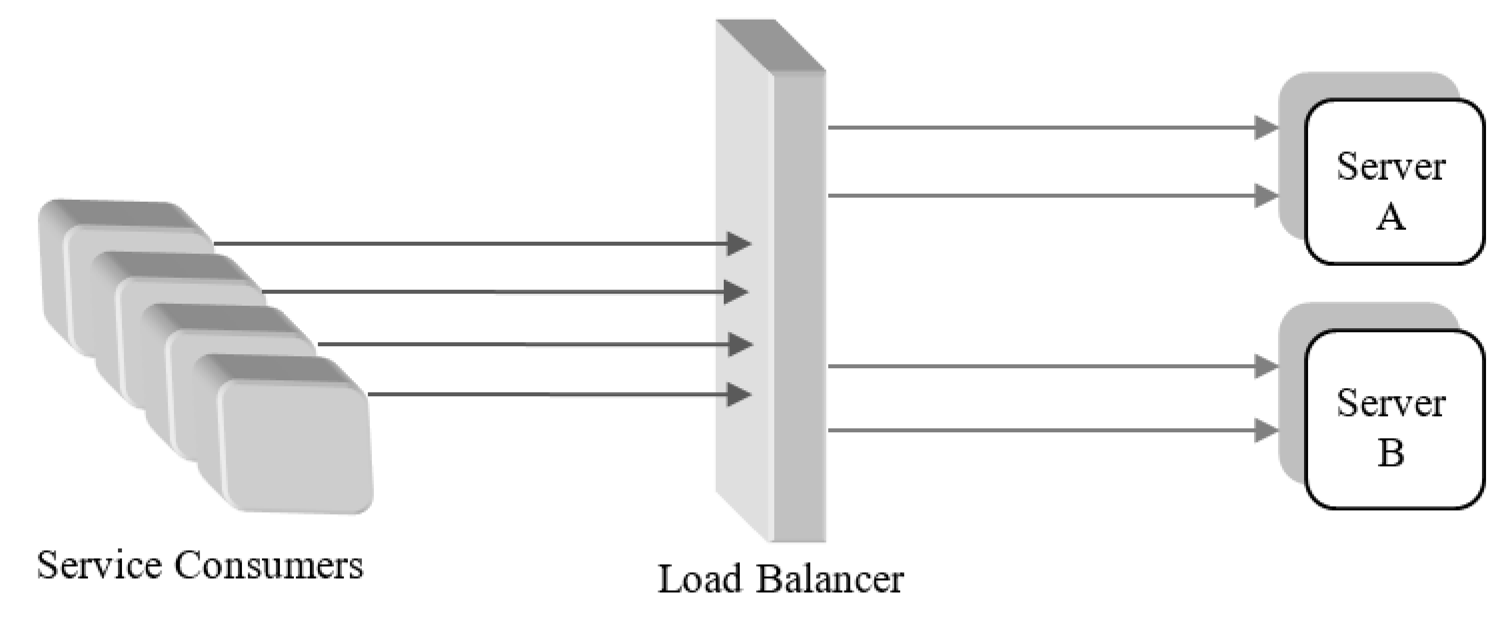
- It allocates loads based on fixed set of pre-configured rules related to the nature of the input traffic.

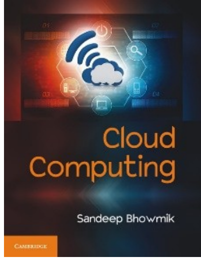
- The scheduling algorithms generally used to schedule the loads among resources, are round robin and weighted round robin scheduling.



Categories of Load Balancing

- **Static Approach**





Categories of Load Balancing

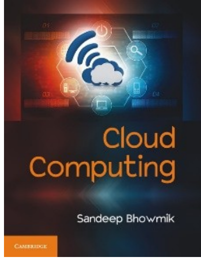
- **Dynamic Approach**

- Load is not distributed among resources depending on any fixed set of rules.

- Scheduling of tasks happen based on current state of the resources.

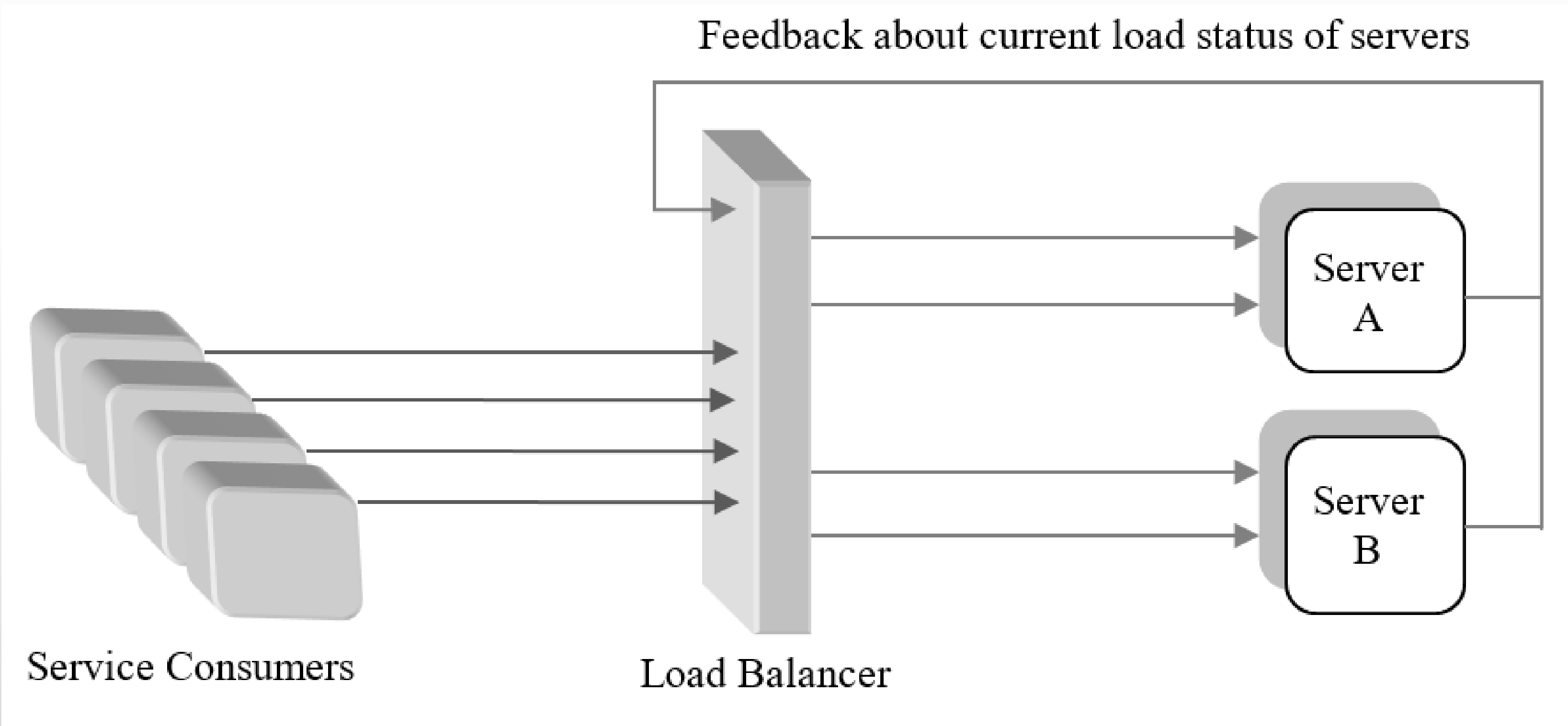
- Through feedback mechanism, resource components are monitored continually.

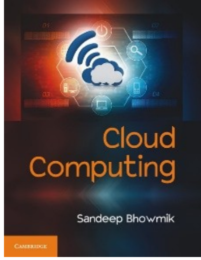
- If any node gets fully loaded or fails, that doesn’t halt the whole system.



Categories of Load Balancing

- **Dynamic Approach**



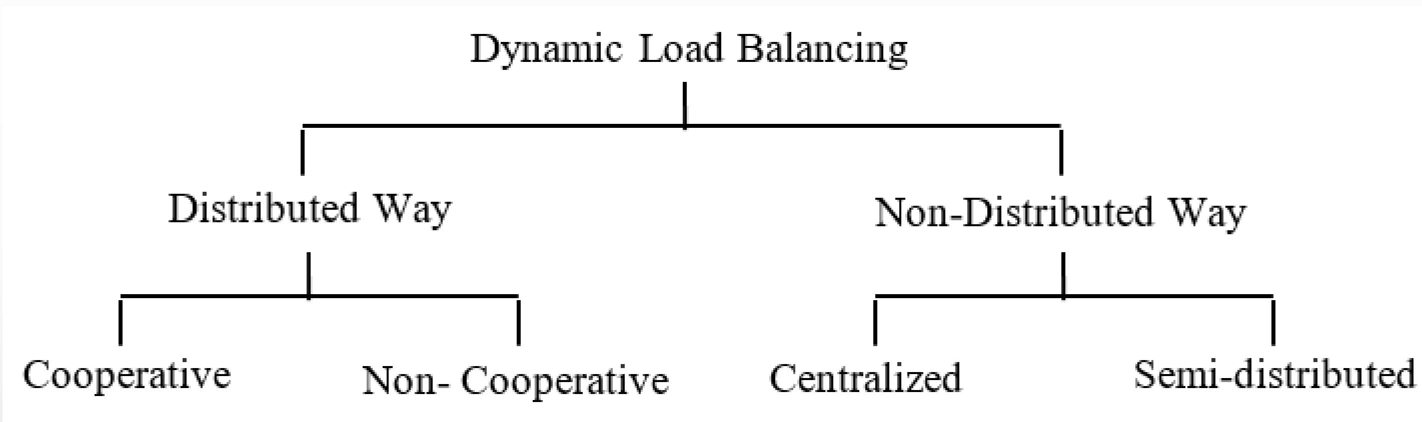


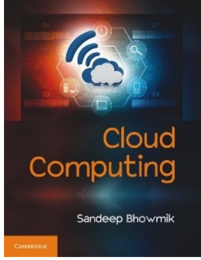
Exploring Dynamic Load Balancing

- The implementation of the technique is done in two separate ways –
    - Distributed way
    - Non-distributed way

- The sharing of the task can again happen in two ways –
    - Cooperative - all the nodes work together in parallel to attain a same goal.
    - Non-cooperative - the nodes work separately to achieve some local goals.

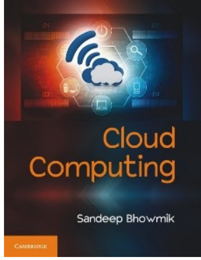
Exploring Dynamic Load Balancing





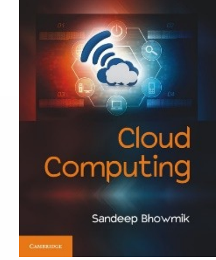
Exploring Dynamic Load Balancing

- In non-distributed dynamic load balancing approach, all the nodes of a distributed system do not take part in balancing system load.
- Here, the task of load balancing is managed by one individual node or a cluster of nodes.
- When it is managed by a single/individual node, the approach is called *centralized* load balancing approach.
- When the load balancing task is managed by a cluster of nodes, the approach is known as *semi-distributed* load balancing.



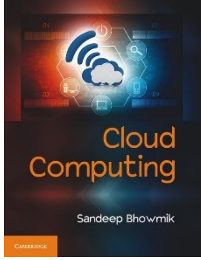
Exploring Dynamic Load Balancing

- The distributed approach –
  - generates more messages than the non-distributed load balancing approach,
  - as there are need of more interaction among the load balancer nodes.
- On the other hand, in distributed approach –
  - multiple load balancers work together,
  - hence, the system does not suffer any major problem even if one or more load balancer nodes in the system fail.



Exploring Dynamic Load Balancing

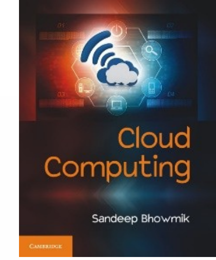
- The centralized approach of load balancing exchanges very few messages to reach a decision.
- Since, the load balancing act of entire system is dependent on a single node that may cause bottleneck problem.
- Distributed load balancing provides more robustness to system at the cost of some added overhead.



Exploring Dynamic Load Balancing

- Several algorithmic solutions have been used by researchers and developers for dynamic load balancing implementation.

- Promising among them are, algorithms suggested based on the theory of
    - Biased Random Sampling,
    - Active Clustering,
    - Honeybee Foraging Behaviour.

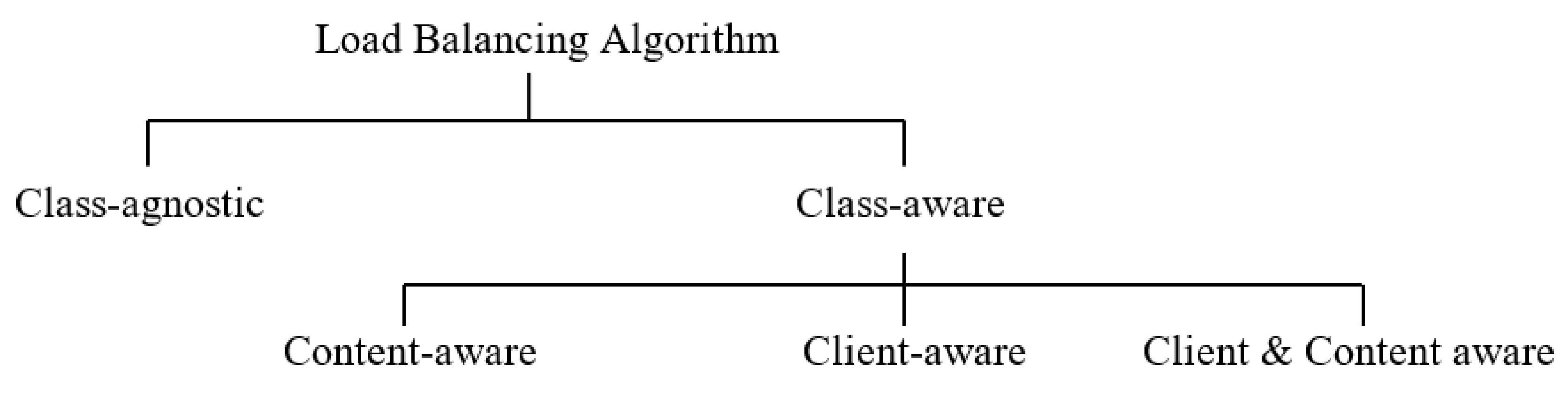


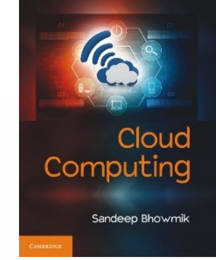
Parameters for Consideration

- Resource Utilization
- Response Time
- System health
- Associated Overhead
- Scalability

Load Balancing Algorithms

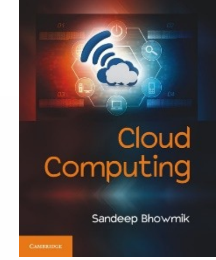
- Classification of load balancing algorithms





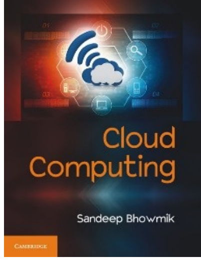
Load Balancing Algorithms

- *Class-agnostic* algorithms make decision without considering the nature and source of incoming requests.
- *Class-aware* algorithms makes decision with knowledge about the nature and source of the service requests.
- Class-aware algorithms are more suitable for balancing loads in critical environment.



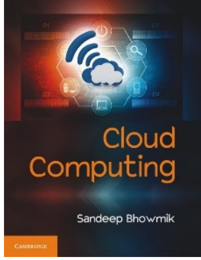
Load Balancing Algorithms

- Depending upon awareness about received content, class-aware algorithms can be classified in two categories.
- When it uses knowledge about the content type of service requests, it is called *content-aware* load balancing.
- When knowledge about content source (that is clients) is used in load distribution, algorithms are known as *client-aware*.
- In practice, applications need to maintain a balance between both the client-aware and content-aware mechanisms.



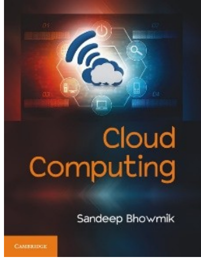
The Persistence Issue

- Load-balancer should uniformly handle a client's requests within a session.
- The system must send all requests arrived within a single session to the same backend server for processing uniformity.
- This character is known as *persistence* or *stickiness*.



The Persistence Issue

- Persistence issue is handled through appropriate tracking of session.
- Session tracking is implemented by storing session data in load balancer.
- Ideally the pool of servers behind the load balancer should be session-aware.
- ADC (Application Delivery Controller) is an advance category of load balancer that enhance the performance of cloud applications.



Case Study

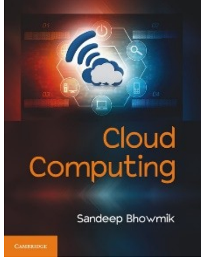
- **Google Cloud**

- Google Compute Engine offers server-side load balancing facility.

- First level of load balancing takes place at the DNS level.

- DNS servers examine and find server cluster which are closer to the origin of the service request.

- At second level, the load balancer assigns the request to suitable server instance in the cluster.



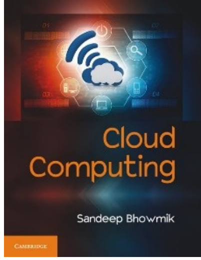
Case Study

- **Amazon EC2**

- The Elastic Load Balancing (ELB) technique at Amazon EC2 has two modules –
  - a set of load balancers
  - a controller service

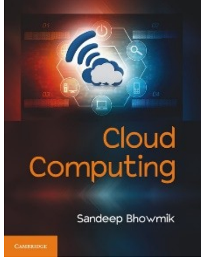
- Here load balancer is configured to link with group of EC2 instances referred as auto scaling group.

- All the servers of that group is load balanced by the respective load balancer.



Case Study

- **Amazon EC2**
- The controller service monitors the load balancers.
- It keeps a constant watch on the health of the load balancers and checks if they are functioning properly.
- If required, the controller service removes or adds more load balancers in the system.



Thank You